

USE CASE SPECIFICATION

Inventory Management System

Document Version: 1.0

Prepared By: *Kalyani Mahajan*

Date: 27/01/2026

Reference Standard: IEEE 830 / IEEE 29148

REVISION HISTORY

Version	Date	Description	Author
1.0	27/01/2026	Initial Use Case Specification	Kalyani

TABLE OF CONTENTS

1. Introduction
 2. Actors
 3. Use Case List
 4. Detailed Use Case Specifications
 5. Use Case Diagrams (Placeholder)
 6. Notes and Assumptions
-

1. INTRODUCTION

1.1 Purpose

This document defines the **use cases** for the Inventory Management System.

It describes system behaviour from the perspective of external users (actors) interacting with the system.

The purpose of this document is to:

- Clearly specify system functionality
- Serve as a reference for implementation and validation
- Complement the Software Requirements Specification (SRS)

1.2 Scope

This Use Case Specification covers:

- Inventory item management
- Stock transaction processing
- Dashboard viewing
- Role-based access behaviour

It applies to Version 1 of the Inventory Management System.

2. ACTORS

2.1 Administrator

Description:

An administrator is a privileged user responsible for managing inventory items, users, and system configuration.

Responsibilities:

- Create and manage inventory items
 - View all transactions
 - Access Django admin panel
 - Monitor system state
-

2.2 Staff User

Description:

A staff user is an operational user responsible for day-to-day inventory operations.

Responsibilities:

- View inventory
 - Create stock IN and OUT transactions
 - Access dashboard
-

2.3 Unauthorized User

Description:

A user who is not authenticated in the system.

Responsibilities:

- None
-

3. USE CASE LIST

Use Case ID	Use Case Name	Primary Actor
UC-01	Login User	Admin / Staff
UC-02	View Inventory Items	Admin / Staff
UC-03	Create Inventory Item	Admin
UC-04	Update Inventory Item	Admin
UC-05	Deactivate Inventory Item	Admin
UC-06	Create Stock IN Transaction	Admin / Staff
UC-07	Create Stock OUT Transaction	Admin / Staff
UC-08	View Dashboard	Admin / Staff
UC-09	View Transactions	Admin
UC-10	Logout User	Admin / Staff

4. DETAILED USE CASE SPECIFICATIONS

USE CASE UC-01: Login User

Primary Actor:

Admin, Staff

Description:

Allows a registered user to authenticate and access the system.

Preconditions:

- User account exists
- User is not already logged in

Postconditions:

- User session is created
- User is redirected based on role

Main Flow:

1. User enters credentials
2. System validates credentials
3. System establishes user session
4. User is redirected to dashboard

Alternate Flows:

- Invalid credentials: error message displayed
- Inactive account: access denied

USE CASE UC-02: View Inventory Items

Primary Actor:

Admin, Staff

Description:

Allows users to view a list of active inventory items.

Preconditions:

- User is authenticated

Postconditions:

- Inventory list is displayed

Main Flow:

1. User navigates to inventory list
2. System retrieves active items
3. System displays items with stock status

Alternate Flows:

- No items available: empty state displayed
-

USE CASE UC-03: Create Inventory Item

Primary Actor:

Administrator

Description:

Allows an administrator to create a new inventory item.

Preconditions:

- User is authenticated as Admin

Postconditions:

- New item is stored in the system

Main Flow:

1. Admin enters item details
2. System validates input
3. System creates item
4. Confirmation message displayed

Alternate Flows:

- Invalid input: error message displayed
-

USE CASE UC-04: Update Inventory Item

Primary Actor:

Administrator

Description:

Allows an administrator to update item metadata.

Preconditions:

- Item exists
- User is authenticated as Admin

Postconditions:

- Item details are updated

Main Flow:

1. Admin selects item
 2. Admin updates allowed fields
 3. System validates changes
 4. System saves updates
-

USE CASE UC-05: Deactivate Inventory Item

Primary Actor:

Administrator

Description:

Allows an administrator to deactivate an inventory item.

Preconditions:

- Item exists
- User is authenticated as Admin

Postconditions:

- Item is marked inactive
- Item no longer appears in active inventory

Main Flow:

1. Admin selects item
 2. Admin chooses deactivate option
 3. System updates item status
-

USE CASE UC-06: Create Stock IN Transaction

Primary Actor:

Admin, Staff

Description:

Allows a user to add stock to an item.

Preconditions:

- Item exists and is active
- User is authenticated

Postconditions:

- Inventory quantity increases
- Transaction record is created

Main Flow:

1. User selects item
 2. User enters quantity
 3. System validates quantity
 4. System records transaction
 5. System updates inventory
-

USE CASE UC-07: Create Stock OUT Transaction

Primary Actor:

Admin, Staff

Description:

Allows a user to remove stock from an item.

Preconditions:

- Item exists and is active
- Sufficient stock available

Postconditions:

- Inventory quantity decreases
- Transaction record is created

Main Flow:

1. User selects item
2. User enters quantity
3. System validates stock availability
4. System records transaction
5. System updates inventory

Alternate Flows:

- Insufficient stock: operation rejected
-

USE CASE UC-08: View Dashboard

Primary Actor:

Admin, Staff

Description:

Displays inventory analytics and summaries.

Preconditions:

- User is authenticated

Postconditions:

- Dashboard data displayed

Main Flow:

1. User accesses dashboard
 2. System retrieves analytics
 3. System displays charts and summaries
-

USE CASE UC-09: View Transactions

Primary Actor:

Administrator

Description:

Allows admin to view complete transaction history.

Preconditions:

- User is authenticated as Admin

Postconditions:

- Transaction list displayed

Main Flow:

1. Admin navigates to transactions
 2. System retrieves transaction records
 3. System displays results
-

USE CASE UC-10: Logout User

Primary Actor:

Admin, Staff

Description:

Ends the user session.

Preconditions:

- User is logged in

Postconditions:

- User session terminated

Main Flow:

1. User selects logout
 2. System ends session
 3. User redirected to login page
-

5. NOTES AND ASSUMPTIONS

- All users must authenticate before access
 - Permissions are enforced server-side
 - Transactions are immutable
 - Inventory quantity is derived from transactions
-